

Felipe Cordera

PhD Candidate, MIT Operations Research Center

fcordera@mit.edu | <https://felipecordera.github.io> | +1 (857) 706-9701

Research Interests: Large-scale optimization for operational systems, particularly under complex downstream effects and incomplete information, with applications in logistics, transportation, and energy.

Education

Massachusetts Institute of Technology

PhD in Operations Research

Advisors: Alexandre Jacquillat & Cynthia Barnhart

Cambridge, MA

Expected May 2027

Universidad de Chile

Master of Engineering Science, Water Resources and Environment

Santiago, Chile

2020

Passed with highest distinction. Thesis: "Stochastic Dual Dynamic Programming Applied to Reservoir Operations: Limitations in Uncertainty Modeling and Presentation of an Efficient Method to Overcome Them."

Universidad de Chile

Professional Degree in Civil Engineering, Water Resources and Environment

Santiago, Chile

2020

Passed with highest distinction; ranked first in the graduating class.

Universidad de Chile

Bachelor of Engineering Science, Civil Engineering

Santiago, Chile

2016

Passed with highest distinction.

Publications

- **F. Cordera**, A. Jacquillat, and S. McDonald. "Optimizing Advanced Air Mobility Operations in a Corridor Network." *INFORMS Optimization Society Conference Refereed Proceedings, 2026. Major revision at INFORMS Journal on Optimization.*
- F. Sepúlveda, D. Alvarado, **F. Cordera**, E. Esperguel, G. Strbac, and R. Moreno. "Cost-Benefit Analysis of Transmission Network Reliability Standards: An Inverse Optimal Power Flow Approach." *IEEE Access*, 2026.
- **F. Cordera**, R. Moreno, and F. Ordóñez. "Unit Commitment Problem with Energy Storage Under Correlated Renewables Uncertainty." *Operations Research*, 71(6), 1960–1977, 2023.
- S. E. Null, M. A. Olivares, **F. Cordera**, and J. R. Lund. "Pareto Optimality and Compromise for Environmental Water Management." *Water Resources Research*, 57(10), e2020WR028296, 2021.

Working Papers

- C. Barnhart, M. Caldara, **F. Cordera**, O. Islam, A. Jacquillat, and A. Schmid. "Order Picking in Large-scale Robotic Mobile Fulfillment Systems."
- R. Bosworth, **F. Cordera**, and A. Jacquillat. "Airport Gate Assignment with Connecting-Passenger Impacts."

Research Talks

- **Order Picking in Large-scale Robotic Mobile Fulfillment Systems.**
 - MIT Center for Transportation and Logistics (CTL) Research Seminar, Massachusetts Institute of Technology, Cambridge, MA, 2026. *Invited talk.*
 - INFORMS Annual Meeting, Atlanta, GA, 2025.
- **Optimizing Advanced Air Mobility Operations in a Corridor Network.**
 - INFORMS Transportation Science and Logistics Society Conference, Massachusetts Institute of Technology, Cambridge, MA, 2026.
 - INFORMS Optimization Society Conference, Atlanta, GA, 2026.
- **Short-Term Operation Scheduling of Renewable Power Systems under Multistage Uncertainty.**
 - LXXXIX Annual Meeting of the Chilean Mathematical Society (SOMACHI), Universidad de O'Higgins, Rancagua, Chile, 2021.

Teaching Experience

Massachusetts Institute of Technology Teaching Assistant, Integer Optimization	Cambridge, MA Spring 2025
Massachusetts Institute of Technology Teaching Assistant, Optimization Methods	Cambridge, MA Fall 2023
Universidad de Chile Teaching Assistant, EL4114 Optimization	Santiago, Chile Fall 2022
Universidad de Chile Teaching Assistant, Water Resources Systems Analysis	Santiago, Chile 2017–2020
Universidad de Chile Teaching Assistant, Environmental Systems Analysis	Santiago, Chile 2017

Industry and Applied Experience

Amazon Robotics Research Scientist II Intern	North Reading, MA May–Aug. 2025
- Developed large-scale optimization methods for task assignment in robotic fulfillment systems, accounting for downstream robot movements and congestion. - Extended and used a high-fidelity discrete-event simulation environment to evaluate optimization methods under realistic fulfillment-center operations.	
Institute for Complex Engineering Systems, Universidad de Chile Project Engineer	Santiago, Chile 2019–2020
Developed optimization methods for medical appointment and patient transportation scheduling for the Chilean Safety Association.	
Faculty of Medicine, Universidad de Chile Project Engineer	Santiago, Chile 2018–2020
- Developed an integer optimization tool for medical degree revalidation scheduling, integrating examination construction and simulated-patient assignment. - Developed a deferred-acceptance-based tool for assigning medical interns to clinical rotations based on their preferences. - Developed an integer optimization model for university course scheduling across interdisciplinary programs.	

Academic Service

- Co-organizer**, MIT Operations Research Center IAP Seminar, “*GenAI Meets OR*,” Massachusetts Institute of Technology, January 2026.
- Conference Volunteer**, INFORMS Transportation Science and Logistics Society Conference, Massachusetts Institute of Technology, 2026.

Honors and Awards

- ENRE Student Best Paper Award**, “Unit Commitment Problem with Energy Storage Under Correlated Renewables Uncertainty,” INFORMS Annual Meeting, 2024.
- Best Graduate Engineer of the 2020 Class**, Chilean Association of Engineers, 2021.
- National Master’s Scholarship**, National Council for Science and Technology (CONICYT, now ANID), Chile, 2017.
- Short-Term Research Program Award**, Universidad de Chile, 2017. Funding for a research visit to Cornell University.
- Pablo Pérez Zañartu Scholarship**, San Carlos de Maipo Foundation, 2016. Awarded to the top incoming student in Civil Engineering, Water Resources and Environment, at Universidad de Chile.
- Dean’s List for Outstanding Academic Performance**, Universidad de Chile, 2011–2013 and 2015.

Technical Skills

Expertise: Large-scale optimization, integer programming, decomposition methods, network optimization and routing, stochastic optimization, discrete-event simulation, machine learning

Programming: Julia, Python, C, MATLAB

Software: Gurobi, FICO Xpress, GAMS

Languages: Spanish (native), English (high proficiency)